# SKILL -1 : GIT VERSION CONTROL

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**Section:2**

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**Tasks:**

1. **Create a new project folder, initialize Git using git init, and configure Git with your username and email.**
   **a. Create a new Project Folder**

```
C:\Users\nagas\OneDrive\Desktop\2025-2026\EvenSem\FSAD\Sec-2>mkdir skill1

C:\Users\nagas\OneDrive\Desktop\2025-2026\EvenSem\FSAD\Sec-2>
```

   **b. Initialize Git**

```
C:\Users\nagas\OneDrive\Desktop\2025-2026\EvenSem\FSAD\Sec-2\skill1>git init

Initialized empty Git repository in C:/Users/nagas/OneDrive/Desktop/2025-202
6/EvenSem/FSAD/Sec-2/skill1/.git/
```

   **c. Configure git with username and email**

```
C:\Users\nagas\OneDrive\Desktop\2025-2026\EvenSem\FSAD\Sec-2\skill1>git config --global user.name "nagasubrahmanyammeka"

C:\Users\nagas\OneDrive\Desktop\2025-2026\EvenSem\FSAD\Sec-2\skill1>git config --global user.email "mekanagasudheeriiitsubbu787@gmail.com"
```

   **d. Verify Configuration**

```
C:\Users\nagas\OneDrive\Desktop\2025-2026\EvenSem\FSAD\Sec-2\skill1>git config --list
diff.astextplain.textconv=astextplain
filter.lfs.clean=git-lfs clean -- %f
filter.lfs.smudge=git-lfs smudge -- %f
filter.lfs.process=git-lfs filter-process
filter.lfs.required=true
http.sslbackend=schannel
core.autocrlf=true
core.fscache=true
core.symlinks=false
pull.rebase=false
credential.helper=manager
credential.https://dev.azure.com.usehttppath=true
init.defaultbranch=master
user.name=nagasubrahmanyammeka
user.email=mekanagasudheeriiitsubbu787@gmail.com
core.repositoryformatversion=0
core.filemode=false
core.bare=false
core.logallrefupdates=true
core.symlinks=false
core.ignorecase=true
```

2. **Create two files (example: main.java, notes.txt) with sample content, check the repository status, and add them to staging.**
   a. **Creating Two files**

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```
C:\Users\nagas\OneDrive\Desktop\2025-2026\EvenSem\FSAD\Sec-2\2400030128_SkillExperiments_SEC2\SKILL-1\skill1>echo. >main.java

C:\Users\nagas\OneDrive\Desktop\2025-2026\EvenSem\FSAD\Sec-2\2400030128_SkillExperiments_SEC2\SKILL-1\skill1>echo .>notes.txt
```

### b. Adding Sample Content

```
C:\Users\nagas\OneDrive\Desktop\2025-2026\EvenSem\FSAD\Sec-2\2400030128_SkillExperiments_SEC2\SKILL-1\skill1>echo "Hello Java" > main.java

C:\Users\nagas\OneDrive\Desktop\2025-2026\EvenSem\FSAD\Sec-2\2400030128_SkillExperiments_SEC2\SKILL-1\skill1>echo "Skill 1 Notes">notes.txt
```

### c. Check Repository Status

```
C:\Users\nagas\OneDrive\Desktop\2025-2026\EvenSem\FSAD\Sec-2\2400030128_SkillExperiments_SEC2\SKILL-1\skill1>git status
On branch master

No commits yet

Untracked files:
  (use "git add <file>..." to include in what will be committed)
        main.java
        notes.txt
```

### d. Add files to Staging

```
C:\Users\nagas\OneDrive\Desktop\2025-2026\EvenSem\FSAD\Sec-2\2400030128_SkillExperiments_SEC2\SKILL-1\skill1>git add .

C:\Users\nagas\OneDrive\Desktop\2025-2026\EvenSem\FSAD\Sec-2\2400030128_SkillExperiments_SEC2\SKILL-1\skill1>
```

### e. Checking status

```
C:\Users\nagas\OneDrive\Desktop\2025-2026\EvenSem\FSAD\Sec-2\2400030128_SkillExperiments_SEC2\SKILL-1\skill1>git status
On branch master

No commits yet

Changes to be committed:
  (use "git rm --cached <file>..." to unstage)
        new file:   main.java
        new file:   notes.txt
```

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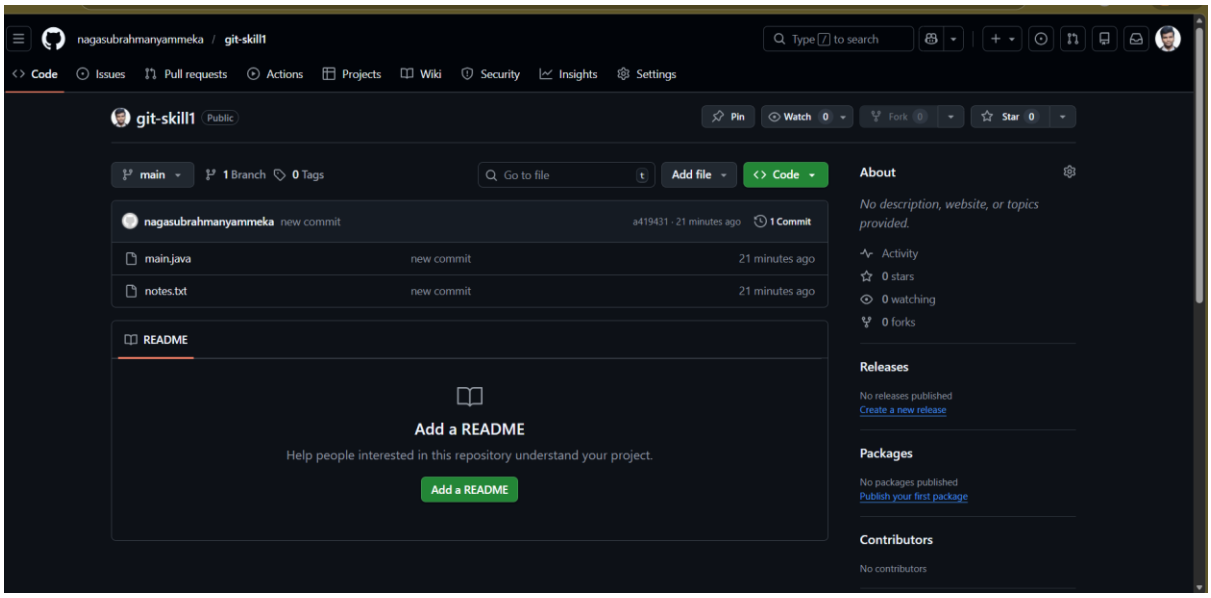
### 3. Commit the staged files with a meaningful commit message

```
C:\Users\nagas\OneDrive\Desktop\2025-2026\EvenSem\FSAD\Sec-2\2400030128_SkillExperiments_SEC2\SKILL-1\skill1>git commit -m "new commit"
[master (root-commit) a419431] new commit
 2 files changed, 2 insertions(+)
 create mode 100644 main.java
 create mode 100644 notes.txt
```

### 4. Created a new repository on GitHub, connect your local repo using the remote URL and push the committed changes.



5. **Created the first branch (example: feature-update), make modifications, and add + commit the changes.**
   a. **Created and Switched to the branch**

```
C:\Users\nagas\OneDrive\Desktop\2025-2026\EvenSem\FSAD\Sec-2\2400030128_SkillExperiments_SEC2\SKILL-1\skill1>git branch -M main
```

   b. **Modified the file**

```
C:\Users\nagas\OneDrive\Desktop\2025-2026\EvenSem\FSAD\Sec-2\2400030128_SkillExperiments_SEC2\SKILL-1\skill1>echo "Feature Update Added" >>main.java
```

   c. **Add and commit**

```
C:\Users\nagas\OneDrive\Desktop\2025-2026\EvenSem\FSAD\Sec-2\2400030128_SkillExperiments_SEC2\SKILL-1\skill1>git add main.java
```

```
C:\Users\nagas\OneDrive\Desktop\2025-2026\EvenSem\FSAD\Sec-2\2400030128_SkillExperiments_SEC2\SKILL-1\skill1>git commit -m "Added Feature Update"
[feature-update 17db9b0] Added Feature Update
 1 file changed, 1 insertion(+)
```

6. **Created a second branch (example: bug-fix), apply changes, and add + commit them.**
   a. **Switched to main**

```
C:\Users\nagas\OneDrive\Desktop\2025-2026\EvenSem\FSAD\Sec-2\2400030128_SkillExperiments_SEC2\SKILL-1\skill1>git checkout main
Switched to branch 'main'
Your branch is up to date with 'origin/main'.
```

   b. **Created new branch**

```
C:\Users\nagas\OneDrive\Desktop\2025-2026\EvenSem\FSAD\Sec-2\2400030128_SkillExperiments_SEC2\SKILL-1\skill1>git checkout -b bug-fix
Switched to a new branch 'bug-fix'
```

   c. **Modified the file**

```
C:\Users\nagas\OneDrive\Desktop\2025-2026\EvenSem\FSAD\Sec-2\2400030128_SkillExperiments_SEC2\SKILL-1\skill1>echo "Bug fixed here" >> main.java
```

   d. **Add and commit**

```
C:\Users\nagas\OneDrive\Desktop\2025-2026\EvenSem\FSAD\Sec-2\2400030128_SkillExperiments_SEC2\SKILL-1\skill1>git add main.java
```

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7. **Switch back to the main branch and merge both branches one after the other**
   a. **Switched to main branch**

```
C:\Users\nagas\OneDrive\Desktop\2025-2026\EvenSem\FSAD\Sec-2\2400030128_SkillExperiments_SEC2\SKILL-1\skill1>git checkout main
Switched to branch 'main'
Your branch is up to date with 'origin/main'.
```

   b. **Merge feature branch**

```
C:\Users\nagas\OneDrive\Desktop\2025-2026\EvenSem\FSAD\Sec-2\2400030128_SkillExperiments_SEC2\SKILL-1\skill1>git merge feature-update
Updating a419431..17db9b0
Fast-forward
 main.java | 1 +
 1 file changed, 1 insertion(+)
```

   c. **Merge bug-fix branch**

```
C:\Users\nagas\OneDrive\Desktop\2025-2026\EvenSem\FSAD\Sec-2\2400030128_SkillExperiments_SEC2\SKILL-1\skill1>git merge bug-fix
Auto-merging main.java
CONFLICT (content): Merge conflict in main.java
Automatic merge failed; fix conflicts and then commit the result.
```

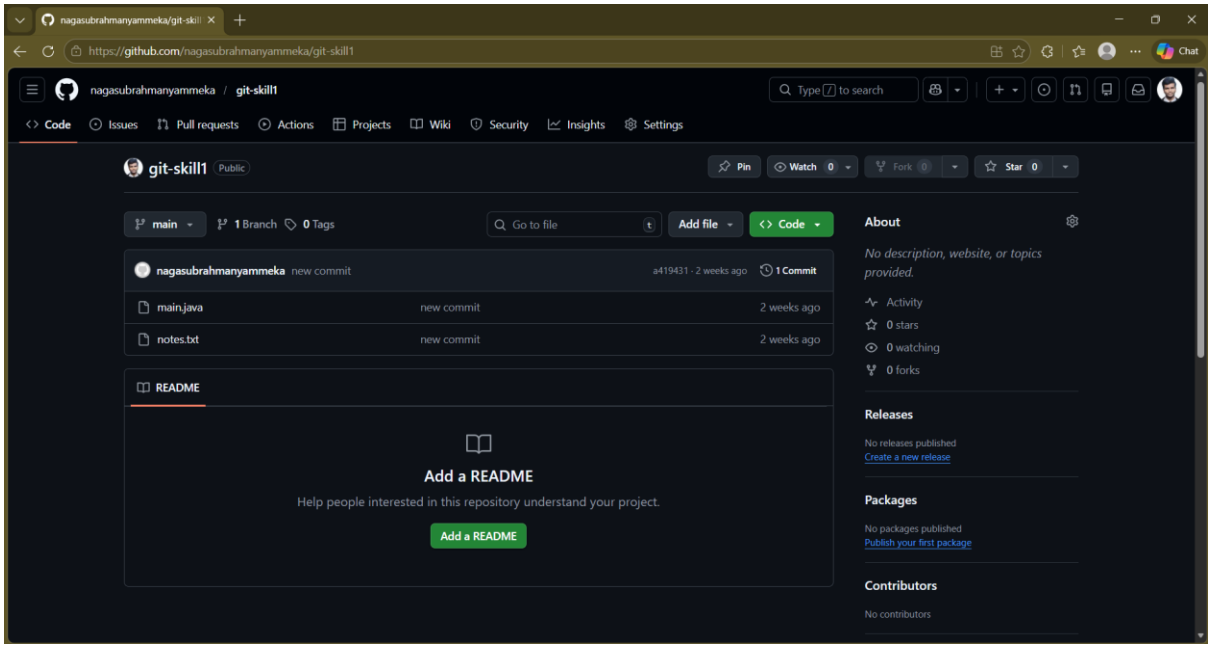
8. **If any merge conflict occurs, resolve it manually, commit the fix, and complete the merge**
   a. **Manually edited the conflicted file**
   b. **Added Resolved file**

```
C:\Users\nagas\OneDrive\Desktop\2025-2026\EvenSem\FSAD\Sec-2\2400030128_SkillExperiments_SEC2\SKILL-1\skill1>git add main.java
```

**Final Log History:**

```
C:\Users\nagas\OneDrive\Desktop\2025-2026\EvenSem\FSAD\Sec-2\2400030128_SkillExperiments_SEC2\SKILL-1\skill1>git log --oneline --graph
*   3801546 (HEAD -> main) Resolved merge Conflict
|\
| * fde3714 (bug-fix) Bug fix applied
* | 17db9b0 (feature-update) Added Feature Update
|/
* a419431 (origin/main) new commit
```

**OUTPUT:**




**GITHUB LINK:**

https://github.com/nagasubrahmanyammeka/SkillExperiments_2400030128_FSAD_Sec2/tree/main/SKILL-1